

Midas

AI Stock Investment Analyst

Framework: AutoGen + Streamlit | Version: 1.0 | Date: June 2026
Repository: bibhu2020/agenticaai (Hugging Face Space)

1. Overview

Midas is an AI-powered stock analysis platform. Given a company name or ticker symbol, a team of four specialist AutoGen agents collaborates to analyse stock trends, recent news, market sentiment, and ultimately deliver an investment decision. The application is built on Microsoft AutoGen and served via Streamlit.

2. System Design

2.1 Components

Component	Technology	Purpose
app.py	Streamlit	UI — ticker input, live agent message stream, icon-tagged chat
teams/investment_team.py	AutoGen RoundRobin GroupChat	Orchestrates the 4-agent analysis pipeline
aagents/stock_trends_agent.py	AutoGen AssistantAgent	Fetches and analyses price history via yfinance
aagents/news_agent.py	AutoGen AssistantAgent	Retrieves and summarises recent company news
aagents/sentiment_agent.py	AutoGen AssistantAgent	Analyses market sentiment from news and social signals

Round	Agent	Tools Used	Output
3	SentimentAgent	Serper / social media signals	Bullish / Bearish sentiment score
4	DecisionAgent	Prior 3 reports (no external tools)	BUY / HOLD / SELL + rationale

4. Termination Conditions

- TextMentionTermination: stops when decision_agent outputs 'TERMINATE'
- MaxMessageTermination: hard cap of 10 messages to prevent runaway loops

5. Configuration

Env Var	Required for
GOOGLE_API_KEY	Default LLM (Gemini 2.0 Flash)
OPENAI_API_KEY	OpenAI model option (gpt-4o-mini)
SERPER_API_KEY	Web search / news tools